

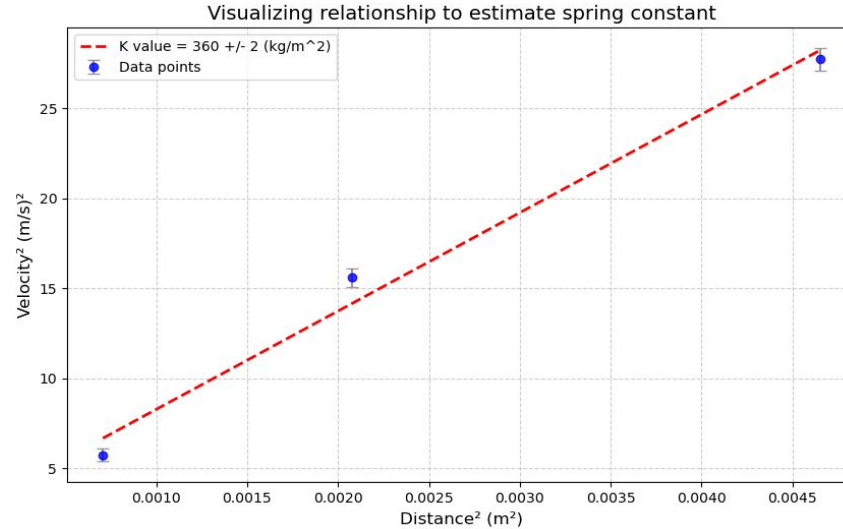
Estimation of Spring constant using velocity

Methods:

Recorded force and displacement for the spring, plotted a best-fit line, and calculated the spring constant by multiplying the slope by the mass. Also measured the moment of inertia and muzzle velocity.

Dominant Source of Uncertainty:

The dominant source of uncertainty is the angle of inclination of the mounted projectile from the horizontal. For the maximum transfer of kinetic energy the muzzle must be mounted perfectly horizontally. If any tilts occur the collision wouldn't be fully elastic.



Findings from this lab:

Muzzle Velocity Calculation: Calculated projectile muzzle velocity at each launcher setting using the average angle and associated uncertainty.

$$V_{\text{short}} = 2.242 \pm 0.35 \text{ m/s}$$

$$V_{\text{medium}} = 3.60 \pm 0.51 \text{ m/s}$$

$$V_{\text{long}} = 5.3 \pm 0.61 \text{ m/s}$$

Spring Constant Calculation: Created a best-fit line for the velocity squared and the distance of the spring compression squared. Then calculated the spring constant by multiplying the slope of the line by the mass of the system.

$$K \text{ Value} = 361 \text{ kg/m}^2 \pm 1 \text{ kg/m}^2$$

Moment of Inertia Measurement:

Measured the pendulum and ball combined mass, and distance to the center of mass to calculate inertia using

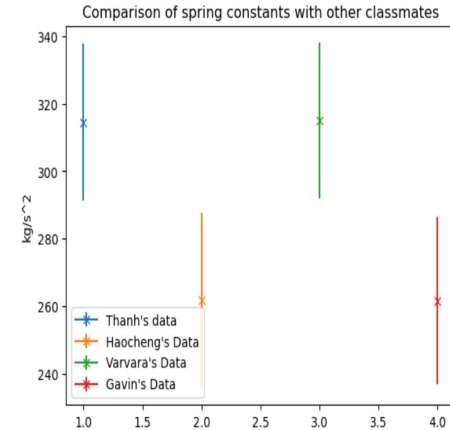
$$I_{\text{simple}} = M \cdot R_{CM}^2 = 0.028 \pm 0.1 \text{ kg} \cdot \text{m}^2$$

$$\text{gram} = 66 \text{ g} \pm 0.1 \text{ g}, \quad R_b = 30 \text{ cm} \pm 1 \text{ cm}$$

Projectile Firing: Fired projectiles (five times per range) and recorded the maximum angle reached by the pendulum with its uncertainty.

The relationship used for calculating the velocity in this lab is

$$v = \frac{1}{mR_b} \sqrt{2IMgR_{cm}(1 - \cos\theta)}.$$



Error Propagation for velocity

$$\Delta v = \sqrt{\left(\frac{\partial v}{\partial t} \Delta t\right)^2 + \left(\frac{\partial v}{\partial M} \Delta M\right)^2 + \left(\frac{\partial v}{\partial \theta} \Delta \theta\right)^2 + \left(\frac{\partial v}{\partial R_m} \Delta R_m\right)^2}$$

$$\Delta v = \sqrt{\left(\frac{\partial v}{\partial t} \Delta t\right)^2} = \Delta t \times \sqrt{\frac{2I}{MR_b} (1 - \cos \frac{\theta}{180})}$$

For More Info

SCAN ME Apparatus



From Music to Measurement: Investigating the RC Circuit Filter Response

INTRODUCTION

Resistance capacitor circuits serve as key components in timing, filtering, and signal shaping. In this lab, we analyze the RC time constant and study how the circuit's output changes with frequency.

THEORY

(~) The voltage ratio for a low-pass filter is given by:

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\sqrt{1 + (2\pi f\tau)^2}}$$

where F is the frequency and τ is the time constant.

(~) The time constant is defined as: $\tau = R \times C$.

(~) At low frequencies, the output voltage is $\approx$ equal to the input.

(~) At higher frequencies, the capacitor's impedance decreases, altering the output.

(~) The observed change in V_{out} / V_{in} versus frequency indicates whether the circuit behaves as a low-pass or high-pass filter.

METHODS

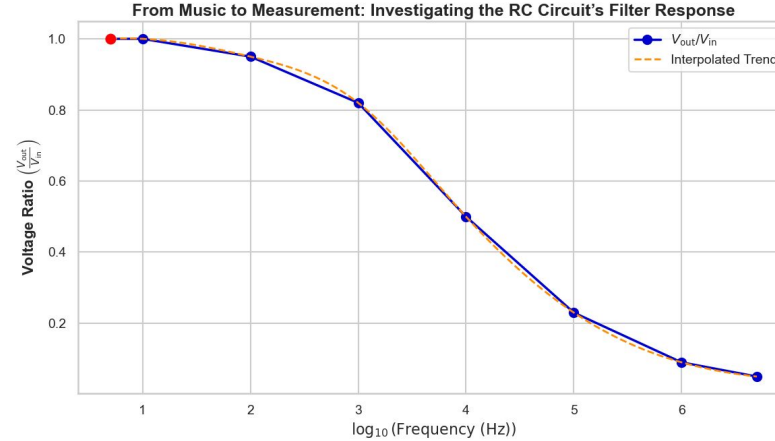
A function generator, oscilloscope, and RC components were used to build the circuit. The theoretical time constant ($\tau = R \times C$) was calculated and compared with oscilloscope measurements. The frequency response was determined by plotting V_{out} / V_{in} versus frequency to identify the threshold behavior.

Dominant Source of Uncertainty

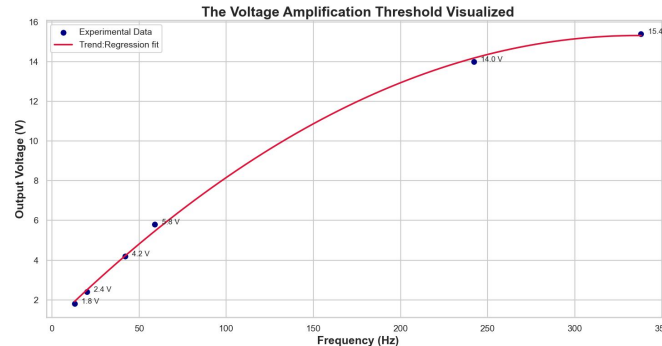
The dominant source of uncertainty in the time constant measurement is the tolerances of the resistor and capacitor. Using uncertainty propagation, the relative uncertainty in the time constant is given by:

$$\frac{\Delta\tau}{\tau} = \sqrt{\left(\frac{\Delta R}{R}\right)^2 + \left(\frac{\Delta C}{C}\right)^2}$$

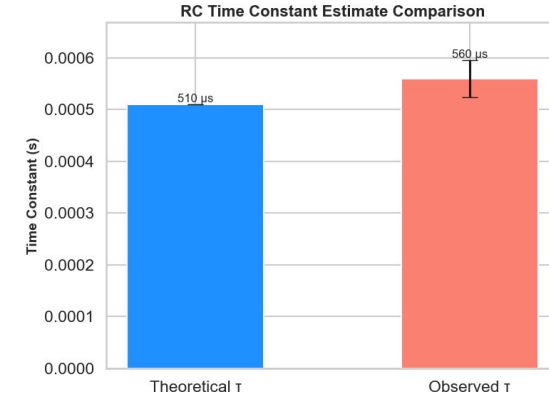
Findings and Results



The theoretical time constant ($510 \mu\text{s}$) ± 36 closely matches the observed value ($560 \mu\text{s}$), validating the RC circuit model. Additionally, the frequency response plot shows that the voltage ratio V_{out} / V_{in} increases from about 0.1 at low frequencies to approximately 0.87 at higher frequencies, indicating that the circuit acts as a high-pass filter.



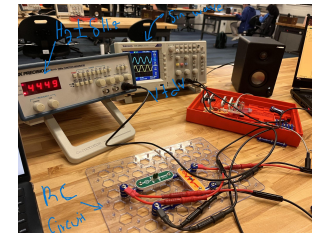
Comparison To Accepted Value



Experimental Reflection

The experiment demonstrated that the RC circuit model not only accurately predicts circuit behavior but also distinguishes between high-pass and low-pass characteristics. The high circuit, which allows higher frequency signals to pass, can be likened to the crisp, clear treble found in modern electronic music, while the low circuit, which attenuates high frequencies, mirrors the deep, resonant bass tones featured in classical and jazz compositions.

Apparatus



Analysis PDF



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